# MCT-454L Mobile Robotics

# Lab Manual #04


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## 1. Screenshots Reference

All screenshots are located in screenshots/ directory of week4/ folder.

## 2. Approach & Observations

This lab focused on using ROS 2 launch files, rosbag recording, and rqt visualization tools along with implementing a follow-the-leader behavior in turtlesim.

A ROS 2 launch file was created to start the turtlesim simulation and teleoperation node simultaneously. A second turtle was spawned using the /spawn service to enable multi-robot interaction.

Rosbag2 was used to record topics including /turtle1/pose, /turtle2/pose, /turtle1/cmd_vel, and /turtle2/cmd_vel. The recorded bag file was replayed successfully to reproduce the turtle movements.

A follow-the-leader node was implemented in Python, where turtle2 subscribes to turtle1's pose and publishes velocity commands to follow it. The follower demonstrated correct behavior with minor approximation due to simple proportional control.

Finally, rqt_plot was used to visualize /turtle1/cmd_vel, showing real-time variations in linear and angular velocity during movement.

## Findings:

- Launch system simplifies multi-node execution
- Rosbag is useful for recording and replaying robot behavior
- Pose-based control enables basic multi-robot coordination
- rqt_plot helps analyze motion commands visually

## 3. Modified Launch File

The file is located in week4/ros2_ws/src/my_launch_pkg/launch/turtlesim_launch.py

## 4. Rosbag Trajectory Data + Brief Analysis

Using the command:

```
ros2 bag record /turtle1/pose /turtle2/pose /turtle1/cmd_vel /turtle2/cmd_vel
```

Trajectory data was extracted from `/turtle1/pose` and `/turtle2/pose`.

**Observations:**

- Turtle1 shows continuous trajectory based on keyboard input
- Turtle2 follows turtle1 with a slight delay
- Path similarity confirms successful follower behavior
- Small tracking error exists due to proportional control

## 5. Follow-The-Leader Code:

```
import rclpy
from rclpy.node import Node
from turtlesim.msg import Pose
from geometry_msgs.msg import Twist
import math

class Follower(Node):

    def __init__(self):
        super().__init__('follower')

        self.sub = self.create_subscription(
            Pose,
            '/turtle1/pose',
            self.update_pose,
            10)

        self.pub = self.create_publisher(
            Twist,
            '/turtle2/cmd_vel',
            10)

    def update_pose(self, msg):
        cmd = Twist()

        # Simple proportional controller
        cmd.linear.x = 1.5 * msg.x
```

```
        cmd.angular.z = 4.0 * msg.theta

        self.pub.publish(cmd)

def main(args=None):
    rclpy.init(args=args)
    node = Follower()
    rclpy.spin(node)
    node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```

## 5. rqt_plot Graph:

The plot is located in week4/screenshots/rqt_plot_graph.png